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Glaciers and ice sheets as a biome

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The tundra is the coldest biome described in typical geography and biology textbooks. Within the cryosphere, there are large expanses of ice in the Antarctic, Arctic and alpine regions that are not regarded as being part of any biome. During the summer, there is significant melt on the surface of glaciers, ice caps and ice shelves, at which point microbial communities become active and play an important role in the cycling of carbon and other elements within the cryosphere. In this review, we suggest that it is time to recognise the cryosphere as one of the biomes of Earth. The cryospheric biome encompasses extreme environments and is typified by truncated food webs dominated by viruses, bacteria, protozoa and algae with distinct biogeographical structures.

A different perspective of icy environments

A biome is commonly defined as a large category of ecosystem characterised by its distinctive association of plants and animals, which in turn are best adapted to the climate, topography and soils of the region [1]. The coldest of the biomes, described in some detail in typical biology and geography undergraduate text books, does not go beyond tundra, leaving most of Greenland, Arctic polar deserts and nearly the whole of Antarctica without a biome characterisation. We argue here that glaciers (see Glossary), ice sheets and the cryosphere in general are a biome uniquely dominated by microorganisms (both autotrophs and heterotrophs) and active biogeochemical processes that have both local and global impacts and should be recognised as such in their own right. Continental Antarctica is largely ignored as a biome, although it has a diverse array of surface lakes, numerous subglacial lakes (e.g. Lake Vostok), ice sheets and some ice free areas, all of which harbour a range of life forms [2–4].

The existence of glaciers and ice sheets, similar to other biomes, is characterised by particular patterns in precipitation, temperature and topographic conditions [5]. Glaciers and ice sheets are dominated by low temperatures associated with precipitation patterns (mostly in the form of snow), in which snow and/or water accumulates as ice in the accumulation zone of glaciers, whereas ice is removed in the summer by melting, sublimation and calving in the ablation zone (Figure 1). The existence, growth and shrinkage of glaciers and ice sheets have occurred over timescales ranging from hundreds of thousands to millions of years. Geographically, glaciers and ice sheets cover circa 15 million km^2, or approximately 10% of the land surface area of the Earth. The extent of ice cover during the last glacial maximum is subject to debate, but was at least three times that of present-day coverage [6]. Thus, compared with other terrestrial biomes, glaciers and ice sheets also have large spatial landscape features. Likewise, they also have a range of well-defined distinct habitats within the icy biome (e.g. surface and subglacial habitats; Box 1).

Biogeochemistry and biodiversity of glaciers and ice sheets

A significant part of the biosphere is permanently cold, with temperatures lower than 5 °C. Microbes dominate these environments and are adapted to function under low temperature conditions. Cold-adapted and cold-loving microorganisms (psychrotolerant or psychrotroph, and psychrophilic, respectively) that are actively metabolising on glaciers and ice sheets are subject to extreme low

Glossary

Ablation zone: the region of glacier where ice loss exceeds gain (Figure 1, main text).
Calving: the breaking off of ice at the snout of a glacier as icebergs.
Cold-adapted enzymes: enzymes that are involved in a range of cellular functions (maintenance and growth) and that have optimal activity at low temperatures.
Cold glaciers: glaciers where the basal temperature is well below freezing point and, thus, are frozen to the bedrock.
Cold-shock proteins: proteins involved in the rapid response to sudden freezing.
Cryoconite holes: small holes at the ice surface filled with dark-coloured inorganic and organic particles (known as cryoconite) and glacial melt. The ice melts because of the lower albedo of cryoconite material relative to the surrounding ice.
Englacial: the area and vains occurring within a glacier.
Exopolymeric substances: substances mostly composed of glycoproteins that are exuded by microbes. They are involved in biofilm formation. In extreme cold habitats, exopolymeric substances give protection against freezing, desiccation and extremely high salinities (in the case of brines in sea ice).
Glacial mass balance: the balance between accumulation (i.e. the zone of a glacier higher up where annual accumulation of ice is higher than melting) and ablation (i.e. the zone of a glacier lower down where annual melting is higher than accumulation). Positive mass balance indicates that accumulation is greater than ablation, whereas a negative balance indicates accumulation is less than ablation.
Glacier: a mass of ice that accumulates from snow and flows slowly downwards.
Ice sheet: a very large mass of ice of continental scale in the shape of a dome flowing outwards. The Antarctic and Greenland Ice Sheets are examples.
Polythermal glaciers: glaciers that have areas of both wet and frozen bases. They occur worldwide.
Psychrophiles: or cold-loving organisms, are characterised by growth at temperatures of approximately 0 °C and the inability to grow at temperatures above 15 °C.
Psychrotrophs and/or psychrotolerants: organisms capable of growth at close to zero temperatures, but they have optimal growth at temperatures higher than 15 °C.
Sublimation: a phase transition of snow and ice directly to vapour. Sublimation plays a major role in ablation from the surface of polar ice.
Temperate or warm-based glaciers: glaciers that have a temperature that is around melting point at their bases as result of friction and geothermal heating. At the base of these glaciers, a well-developed wet system can develop (subglacial water) that may be augmented by supraglacial drainage.

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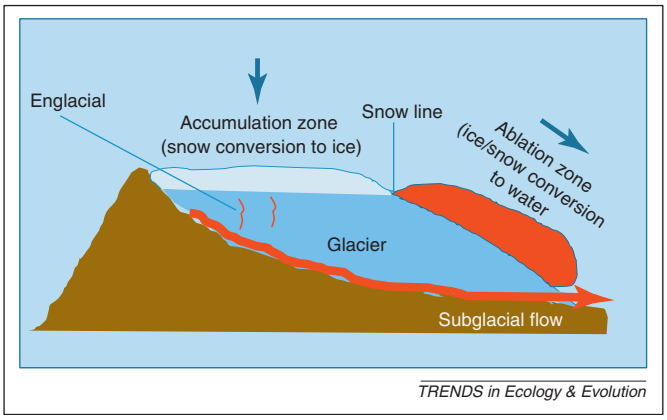

Figure 1. Simplified mass fluxes in an idealised small valley glacier.

temperatures, desiccation and often nutrient limitation. Such organisms have a range of adaptations related to cell metabolism processes [7]. For example, they produce cold-active enzymes, anti-freezing proteins and exopolymeric substances that provide cell protection against the potentially damaging effects of low temperatures [8–10]. Maintaining the fluidity of cell membranes is crucial at low temperatures. This is achieved by an increase in the proportion of unsaturated fatty acids and a decrease in the average fatty acid chain length [11]. Investigations of glacial habitats have revealed that microbial communities are active in biogeochemical transformations such as carbon fixation and respiration, iron cycling and methanogenesis, with important implications at local and global scales [12–15].

Glacial surfaces

During the short melting season, the presence of water provides habitats in which several biogeochemical processes can take place at the surface of the ice and snow, particularly when microbial communities are associated with the sediments that cover substantial areas of the ablation zone of glaciers and ice sheets [12]. For example, rates of photosynthesis and respiration are of the order of several μg of carbon per gram of sediment per day on the surface of Svalbard glaciers and the Greenland Ice Sheet during the summer, which is comparable to microbial activity in sediments and soils in temperate latitudes [14,16]. The organic carbon concentration of the debris at the surface of glaciers and ice sheets is surprisingly high (typically approximately 2–5% [17]) compared with the usual organic carbon concentration of debris from subglacial habitats and the marginal moraines (typically approximately 0.5% [18]). There is now good evidence that microbial activity on the surface of glaciers is important in either producing organic carbon via the positive net balance between photosynthesis and respiration [14] and/or by transforming and trapping allochthonously (externally) derived organic matter via the production of exopolymeric substances [16]. Organic carbon produced and/or transformed at the surface of the ice may be redistributed, providing a labile source of carbon to adjacent terrestrial habitats surrounding glaciers and into coastal waters via glacial runoff [15,19]. Direct evidence of nitrogen fixation on the surface of glaciers has also recently been demonstrated [20]. These authors showed that nitrogen fixation is particularly high during the later stages of the ablation season, when nitrogen sources in the ablation zone become depleted.

Surface microorganisms are derived from several different habitats [17], but it is evident that distinct communities develop and that these comprise species that are tolerant of low temperatures and able to overcome seasonal desiccation [21,22]. Using 16S rRNA analyses, Edwards *et al.* [17] produced the first detailed comparison of bacterial diversity in cryoconite holes on neighbouring glaciers in Svalbard. The clone libraries confirmed the presence of at least seven bacterial phyla within cryoconite sediments and the dominance of the classes Alpha- and Betaproteobacteria is in agreement with earlier culture-dependent studies in alpine glaciers [23]. Cyanobacteria are also strongly represented in cryoconite holes. Large differences

Box 1. The different habitats of the glacial biome

The ice surface

The surface of glaciers receives large amounts of fine wind-blown particles that are cemented together into aggregates by microbial exudates [12,16]. The dark particles melt into the ice to form cryoconite holes. These are effectively mini-lakes (between a few centimetres to approximately 1 m in width and up to 1 m in depth) with the cryoconite sediment overlain by water, which cover between 1% and 10% of the ablation zone of glaciers. Cryoconite also provides sediment in supraglacial streams and lakes. Ice shelves that abut the Antarctic continent and the high Arctic land-masses support seasonal lakes. Similar to glaciers, they may have significant surface sediment that lines the lake basins. Some of these lakes are ephemeral, whereas others persist from year to year and are characterised by well-developed cyanobacterial mats [69].

Subglacial environments

Water can be present at the beds of many glaciers and ice sheets because a small amount of geothermal heat rising from the bedrock is often sufficient to melt and maintain a thin water layer at the interface between the rock and basal ice [70]. The beds of smaller warm-based glaciers may be fed by surface meltwater, which flows in large supraglacial channels until they incept crevasses or large holes called moulins, which extend from the glacier surface to its bed [5]. Beneath the Antarctic ice sheet, which is up to 4 km thick in places, there are numerous subglacial lakes. At least 30 000 km^2 of the ice sheet is likely to be underlain by such lakes [71].

Englacial habitats

The perennial ice of the Dry Valley lakes in Antarctica can be up to 6 m thick and may be heavily laden with debris. Liquid water is present within the ice for up to 150 days a year and may contribute up to 40% of the total ice cover volume [4]. The annual ice of alpine lakes develops significant slush layers that support a complex microbial community [43,44]. However, there is still much to learn about annual ice covers in the Arctic and Antarctic. Glacial ice and ice sheets also provide a habitat for bacteria in aqueous veins extending along lines where ice crystals meet, forming an interconnecting network.

Snow

Snow falls on the polar regions and seasonally on lower latitude biomes. It too forms an important part of the cryosphere, providing a habitat for microbial activity and having an impact on the soil habitats that it covers.

in microbial diversity between snow samples and meltwater from two different sampling periods on a Svalbard glacier suggest that the structure of the microbial community evolves during the melt season [24].

Subglacial habitats

Liquid water also occurs at the beds of temperate and polythermal glaciers, as well as large sectors of polar ice sheets. These habitats contrast with glacial surface environments, having higher rock:water ratios, higher contact times between the bedrock and water, and a lack of light and redox potential that tend towards anoxic conditions [25]. Such environments are the domain of heterotrophic and chemoautotrophic prokaryotes, as well as many fungal species. There is a growing database of the molecular makeup of the communities that live at the base of glaciers, but there is little information on their role in biogeochemical cycling. However, one can speculate that microbial reactions involve oxidation and reduction of several different compounds. Such reactions are likely to be similar to those that occur in river and lake sediments. There is some evidence to indicate that microbial oxidation of bedrock organic matter occurs [13]. For instance, geochemical signatures of subglacial runoff indicate the microbial oxidation of sulphide minerals and organic carbon in bedrock essentially drive these environments towards anoxia, designating them as potential sites of anaerobic processes, including methanogenesis [13,26]. Methanogens have been detected and cultured from the bases of a wide range of glaciers and ice sheets, and may convert significant amounts of carbon to methane over glacial–interglacial cycles, with the potential to impact atmospheric methane concentrations during periods of glacial retreat [13,27]. Molecular analysis of subglacial environments in New Zealand, Alaska and Canada revealed that the class Betaproteobacteria represented a major component of the community, but there were differences in diversity between glaciers that may be related to the nature of the physicochemical environment [28,29]. Butinar *et al.* [30] isolated 22 species of yeast from subglacial environments in Svalbard; many were psychrotolerant and able to use a wide range of carbon and nitrogen sources. High activity under *in situ* conditions suggested that yeasts could be important subglacial drivers of anoxia, because they consume oxygen during respiration, and could also be important in producing substrates involved in methanogenesis.

Subglacial lake systems

Beneath the Antarctic ice sheet, there are 350 recorded subglacial lakes [31]. Some 50% of the bed underlying the Antarctic ice sheet is thought to be warm based, so there is potential for the existence of an extensive drainage system [32]. The best known of these lakes is Lake Vostok (250 km in length, up to 50 km wide and >500 m deep). To date, none of these lakes have been penetrated, so one can only speculate about life in lakes that have been separated from the atmosphere for millions of years. Accretion ice, derived from the lake waters at the base of the ice core above Lake Vostok, provides some indication of the microbial communities that might be found. Molecular analysis revealed members of the classes Alpha-, Beta- and Gammaproteobacteria (Phylum Proteobacteria), the phyla Firmicutes and Actinobacteria, and the genus *Bacteriodes* [33–35].

Life within ice

Biological activity also occurs inside ice, as analysis of ice cores has demonstrated [36]. Knowledge of englacial life within ice is largely based on analysis of ice cores derived from the Greenland Ice Sheet Project (GISP). Molecular analysis has revealed that bacteria are present, but there is only limited information on their functioning. It appears they have extremely low metabolic rates, using energy to repair damaged DNA and amino acids, but do not grow and reproduce [37]. Isolates from the GISP2 cores at depths of 3042.67 m and 3042.80 m contained direct counts of cell concentrations from 6.1×10^7 to 9.1×10^7 cells ml^{-1} of melted ice [38]. Morphotypes belonging to the phyla Firmicutes, Actinobacteria and Proteobacteria were present. The genera *Arthrobacter* and *Microbacterium* (Phylum Actinobacteria) constituted 14 of the 17 isolates. One isolate affiliated with the genus *Paenobacillus* (Phylum Firmicutes). Anaerobic cultures revealed Alpha-, Beta- and Gammaproteobacteria, as well as relatives from the phylum Bacteriodes and the genera *Thermus* and *Clostridium*. Many of the bacteria sequences were closely related to known organisms, but there were new sequences possibly representing new genera and species [39,40].

Ice cores from locations worldwide provide a broader perspective of bacterial communities within ice. When samples were grown on microbiological plates, there were significantly higher numbers of individual colonies or colony forming units (CFU) in 200-year-old ice from the Guliya Ice Cap in Asia (180 CFU ml^{-1}) compared with 1800-year-old ice from the Taylor Dome in Antarctica (10 CFU ml^{-1}), and ice of similar vintage from the Antarctic Peninsula and Greenland had even lower CFUs. Some of these differences relate to non-polar glaciers being closer to major sources of airborne particles from soils [41].

Lake ice cover

Ice covering polar and alpine lakes varies enormously, from the debris-rich perennial ice of the McMurdo Dry Valleys in Antarctic, through glassy annual ice in the Arctic and Antarctic to the slushy topped ice-covers of alpine and some sub-Arctic and Arctic lakes. These offer different types of environment supporting distinct microbial communities. The perennial ice communities of the McMurdo Dry Valley lakes are dominated by cyanobacteria and complex microbial consortia within the ice are effectively enriched microzones where carbon and nitrogen fixation by autotrophs support carbon cycling [42]. Alpine lake ice cover formation involves significant surface flooding from the underlying waters. The upper slushy layer and underlying layers develop complex microbial communities of bacteria, algae, protozoa and rotifers [43,44].

Snow can also provide a habitat for microbial activity. So-called 'red snow' has been recorded since ancient times and is formed by dense populations of phytoflagellates. There are many species, for example *Chlamydomonas nivalis, Raphidonema nivale* and *Chloromonas nivalis* [45–47]. These autotrophs carry secondary carotenoids, astaxanthin and its fatty ester derivatives that confer the characteristic

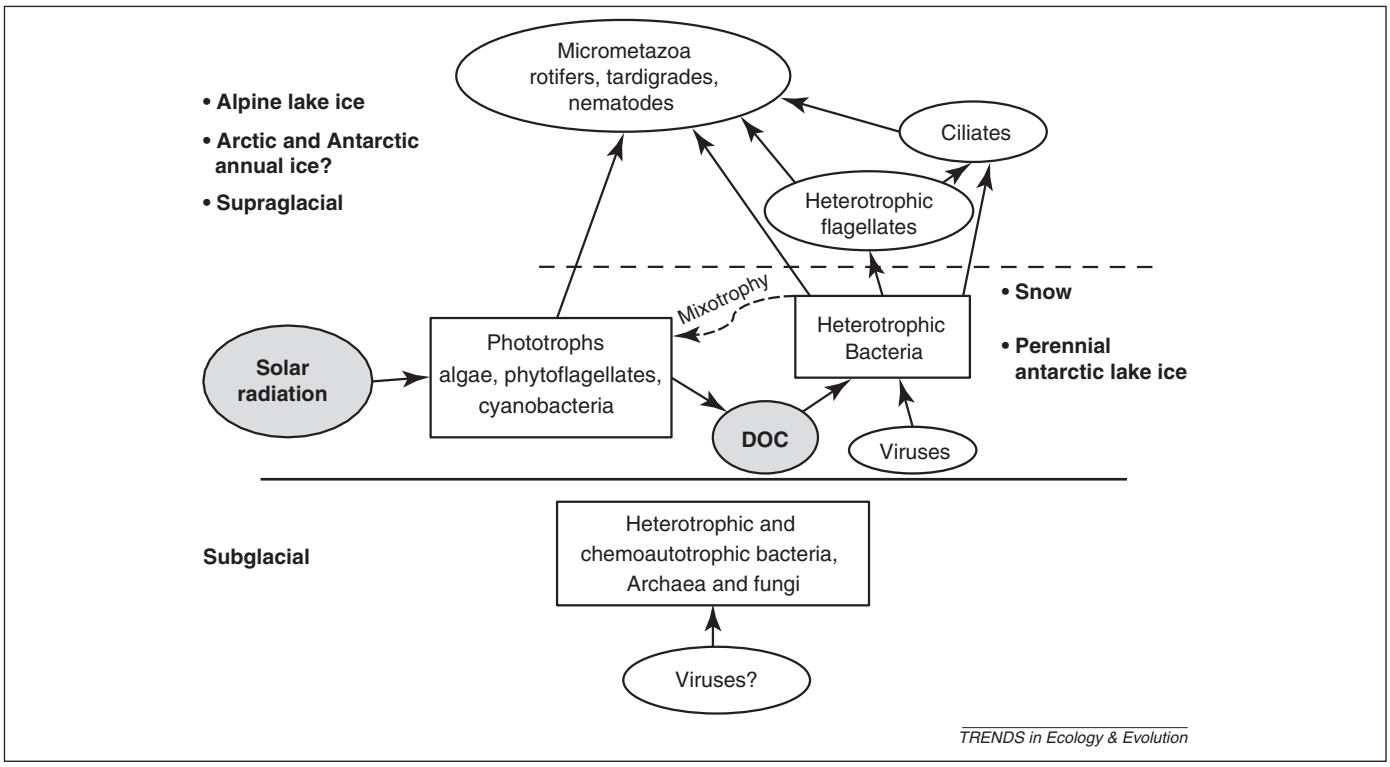

Figure 2. Microbial food webs in different cryospheric environments. Bold text indicates the different habitats. Phototrophs exude part of their photosynthate as dissolved organic carbon (DOC) that provides an energy source to heterotrophic bacteria. Some autotrophs are mixotrophic feeding on bacteria and DOC as well as undertaking photosynthesis. Viral infection and lysis of bacteria short-circuits the transfer of carbon up the food web, recycling carbon and other nutrients to the pool.

colouration to the snow surface [48]. There is limited information available about life within the snowpack, but bacteria occur, even at the South Pole, where measurable production has been recorded [49]. In the Arctic, European Alps and Tateyama Mountains (Japan), bacterial concentrations range from $1.1 \times 10^4\,\mathrm{ml}^{-1}$ to $2.3 \times 10^5\,\mathrm{ml}^{-1}$ [50–52].

Glacier and ice sheet biogeography

Biogeography is defined as the study of the distribution of animals and plants in space and time and is widely used to characterise the different biomes on Earth. The distribution of animals and plants is shaped by events such as the interactions of an organism with its environment (e.g. influence of temperature, light availability and liquid water), biological interactions (e.g. competition and grazing), migration, speciation and extinction. These concepts can also be applied to microbial communities [53], although the relative role of migration and dispersal in microbial communities is considered to be relatively dominant, because microbes are abundant and very small. Both cold-adapted microorganisms and species with cosmopolitan distributions have been identified in a range of polar habitats, suggesting that the distribution of microbial species in the cryosphere is the outcome of a combination of two opposing theories in microbial ecology: endemism and cosmopolitanism. Although many of the species that have colonised polar freshwaters are found in warmer regions, others are restricted to cold environments but may be found in both polar regions. For example cyanobacteria species from a range of habitats in the high Arctic proved to have 99.0% sequence similarity to species previously identified as being endemic to Antarctica [54]. By contrast, in ciliated protozoa, there is low congruence between the high Arctic and the Antarctic species [55]. Even polar oceans (where fewer dispersal barriers are found relative to terrestrial environments) contain specific molecular diversity that seems to be restricted to cold environments [56]. Although microbes are easily dispersed by air, their long-term survival in the atmosphere requires distinct adaptations (e.g. resistance to high UV radiation, extreme cold and desiccation); similar to the adaptations that enable microorganisms to survive in the icy environments [57]. Unsurprisingly the atmosphere has recently been considered to have a distinct microbial biogeography [58]. Although both distinct biogeographical distributions and examples of cosmopolitanism are found in glaciers and ice sheets around the globe [17,21,26,59], the overall microbial community composition in the icy biome seems to be determined by non-random factors.

As is the case for other biomes, we can identify specific biological interactions, adaptations and combinations of biogeochemical process that are unique to icy ecosystems. Among biomes, glaciers and ice sheets are uniquely dominated by microorganisms. The cryospheric biome encompasses extreme environments and, similar to all extreme environments, such as Antarctic lakes, it is typified by truncated food webs (but not necessarily with low diversity) dominated by viruses, bacteria, algae, fungi and protozoa, with few Metazoa (Figure 2). These food web structures do not represent an initial stage of succession, but are likely to

Box 2. The glacial and ice sheet biome

Figure I provides a description of the glacial and ice sheet biome that is similar to the descriptions used in undergraduate text books (e.g. [1]) to describe terrestrial biomes. The description is based on location, annual temperature profile, biological activity and community composition.

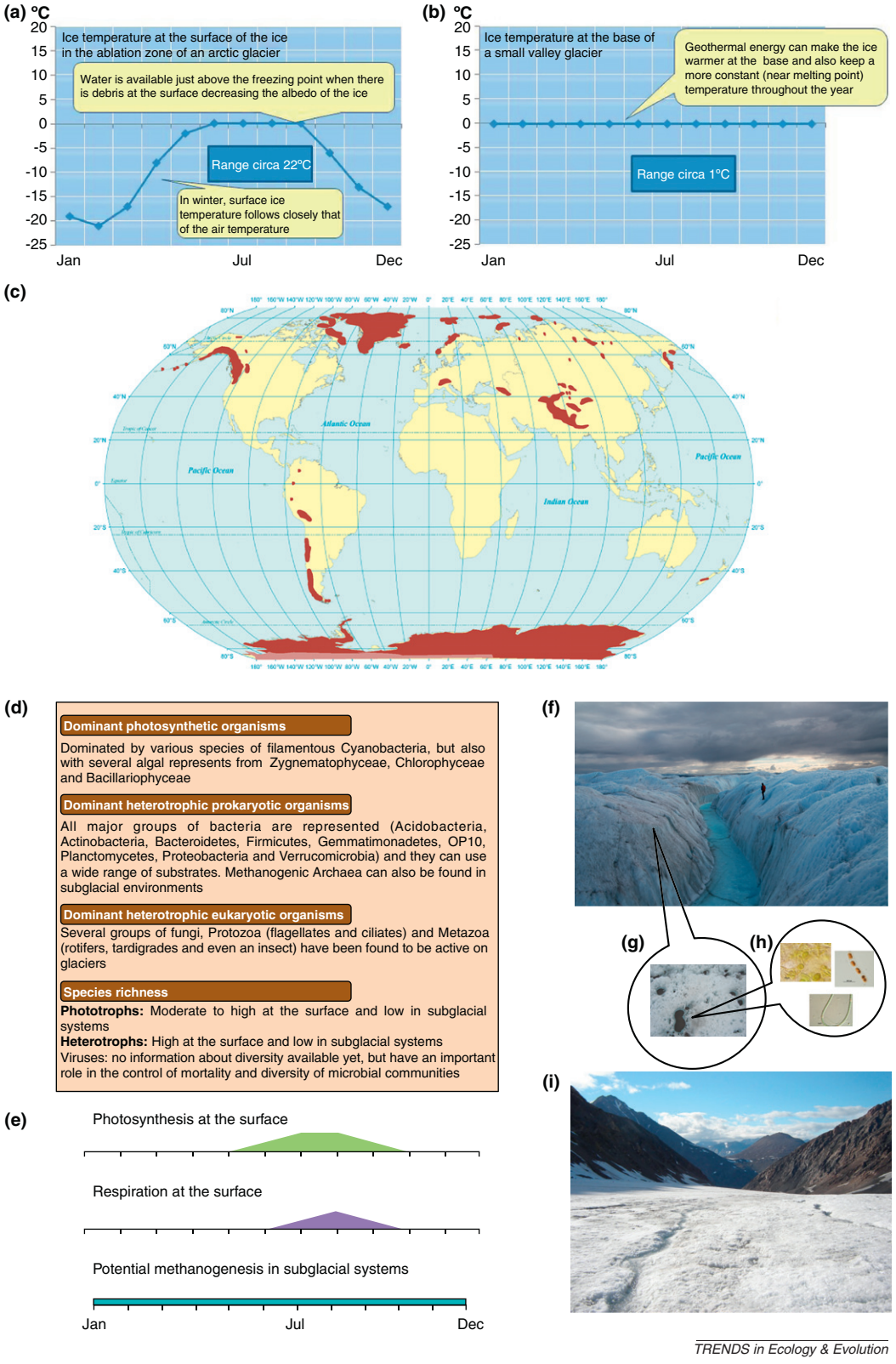

Figure I. The icy biome features. Annual patterns in temperature at the surface of the ice **(a)** and in a typical subglacial system **(b)** using the Russel glacier in Kangerlussuaq, Greenland (63°N) as a model. **(c)** Approximate global distribution of glaciers and ice sheets adapted from GLIMS project (Global Land Ice Measurements from Space) [72]. **(d)** Typical microbial community composition in glaciers and ice sheets. **(e)** Hypothetical annual biological activity for some key microbial processes. The Greenland Ice Sheet (point 660) near Kangerlussuaq **(f)**, cryoconite holes at the surface of the ice in Frøya glacier in Greenland **(g)** and typical algal species inhabiting the ice and cryoconite holes **(h)**. **(i)** Typical valley glacier on the east coast of Greenland (near Zackenberg Station). Adapted from [73] **(d)**. Photographs reproduced, with permission, from Chris Bellas **(f)**, Birgit Sattler **(g** and **i)** and Marian Yallop **(h)**.

represent mature and stable microbial communities with distinct microbial diversity operating within a stable, long timescale physicochemical framework. Many extreme environments (e.g. hot springs and acid mine drainage) fall into this category and, similar to glaciers and ice sheets, are often cited as analogues for life on other planets and early life on Earth. The global importance of glaciers and ice sheets as part of the land mass of Earth suggests that these ecosystems should be regarded as a distinct biome classification (Box 2). The high primary productivity on the surface of glaciers that contributes labile organic matter and nutrients to adjacent environments, and anoxic subglacial methane production and potential release, are just a few examples of how glaciers and ice sheets contribute to local and global biogeochemical cycles.

Concluding remarks and future perspectives

Viruses, bacteria, and protists exhibit high diversity in polar regions that contrasts with a marked decrease of biological diversity among fauna and flora with increasing latitude. The reason for this could be that the mechanisms for microbial dispersal are much greater than in macro plants and animals, and the fact that most of the biosphere of the Earth is cold and a substantial representation of microbial species are adapted to surviving and functioning under low temperature conditions. Among microorganisms, the potential for evolutionary adaptation is considerably enhanced by the phenomenon of horizontal gene transfer (HGT). This is the transfer of genes between related and unrelated organisms mediated by viruses, plasmids and other elements, and has been proposed to be potentially high in low temperature habitats, possibly contributing to higher than expected microbial diversity in polar regions [60].

Only a few studies have addressed the biodiversity and functionality of glacial habitats, mostly based on microscopic observations [61,62], culture-dependent [23,28,63] and PCR-based approaches [17,26,29,64]. At present, relatively little is known about the functional diversity of glacial microbes, and their role in biogeochemical processes. Trait expression in addition to genetic diversity is valuable in explaining microbial adaptation to a habitat and possible responses to climate change [65]. Future investigations of metagenomes and transcriptomes will enable further understanding of the genetic and functional characteristics of the proposed cryosphere biome. Further studies are also necessary to understand the activity of microbial communities in snow and englacial habitats.

The inclusion of glaciers and ice sheets as a biome with unique life adaptations has implications for the conservation of these climate-sensitive ecosystems. Worldwide, glaciers and ice sheets are retreating, particularly in the Alpine and Arctic regions [66], and their biogeochemical links with neighbouring ecosystems via meltwater export need to be understood. During the Snowball Earth, glaciers and ice sheets represented the dominant biome on the planet [67]; consequently, understanding microbial life cycles in extreme cold conditions can inform understanding of the evolution and spread of life on Earth. As glaciers and ice sheets retreat and give way to the development of tundra, unique food web interactions will be compromised [68]. Furthermore, of most concern will be the loss of biodiversity and potentially the loss from the biosphere of a pool of genes adapted to surviving and thriving in the cold.

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